

# Caifen Liu

PhD Candidate in Epidemiology and Biostatistics

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## EDUCATION

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**The University of Hong Kong**

Hong Kong SAR, China

*PhD in Epidemiology and Biostatistics*

Sep 2022–Sep 2026 (expected)

Thesis: *Estimation and Interpretation of COVID-19 Severity and Vaccine Effectiveness.*

Supervisor: [Prof. Ben Cowling](#).

**Southern University of Science and Technology**

Shenzhen, China

*MSc in Statistics*

Sep 2018–Jun 2020

Thesis: *Bivariate Binomial Distribution for Comparing Binomial Observations in Two Correlated Groups.*

Supervisor: [Prof. Guo-Liang Tian](#).

**Hunan University**

Changsha, China

*BE in Materials Science and Engineering*

Sep 2014–Jun 2018

## RESEARCH INTERESTS

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I am interested in developing and applying causal inference methods to observational health data to reduce bias, strengthen the validity of causal conclusions, and support clinical decision-making.

**Methodology:** causal inference; causal estimands; selection bias; inverse-probability weighting; survival and longitudinal data analysis; simulation and sensitivity analyses; missing data methods; target trial emulation.

**Applications:** observational studies using linked electronic health records, registries, and cohort data; vaccine effectiveness evaluation; disease severity evaluation; pharmacoepidemiology.

## DOCTORAL RESEARCH

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**The University of Hong Kong** *School of Public Health*

2022–Present

**COVID-19 severity pyramid and in-hospital progression**

Investigated how changing admission practices altered COVID-19 severity estimates and longitudinal in-hospital progression. Modeled about 2.8 million linked case records, including about 120,000 hospitalizations, using Bayesian severity-pyramid, Aalen–Johansen competing-risk, and Cox models.

**Hospital pressure, admission selection, and fatality risk**

Assessed how hospital pressure, admission selection, and admission severity shaped the observed in-hospital fatality. Applied period-specific generalized additive models and model-based counterfactual standardization to about 120,000 hospitalizations.

**Causal estimands and selection bias in test-negative designs**

Examined when vaccine-induced severity attenuation drives test-negative estimates away from source-population causal estimands. Used causal DAGs, analytic bias derivations, Monte Carlo simulation, and inverse-probability weighting to evaluate bias and correction strategies.

**Hospital-based test-negative design estimates of vaccine effectiveness** (*ongoing*)

Uses simulation to evaluate when hospital-based TNDs recover vaccine effectiveness against target-pathogen hospitalization, benchmarked against randomized-trial and adjusted-cohort estimates under alternative hospital eligibility, testing, and control-selection mechanisms.

### Peer-Reviewed Publications

†Co-first author

1. **Liu, C.**, Okoli, G. N., Chen, R., Sullivan, S. G., and Cowling, B. J. (2026). SARS-CoV-2 vaccination and attenuation of breakthrough infection severity: A systematic global review and meta-analysis. *Clinical Infectious Diseases*, ciag346.
2. Du, Z., **Liu, C.**<sup>†</sup>, Bai, Y., Wang, L., Lim, W. W., Lau, E. H. Y., and Cowling, B. J. (2024). Predicting efficacies of fractional doses of vaccines by using neutralizing antibody levels. *JMIR Public Health and Surveillance*, 10, e49812.
3. Du, Z., **Liu, C.**<sup>†</sup>, Wang, C., Xu, L., Xu, M., Wang, L., Bai, Y., Xu, X., Lau, E. H. Y., Wu, P., and Cowling, B. J. (2022). Reproduction numbers of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) variants: A systematic review and meta-analysis. *Clinical Infectious Diseases*, 75(1), e293–e295.
4. Du, Z., Wang, C., **Liu, C.**<sup>†</sup>, Bai, Y., Pei, S., Adam, D. C., Wang, L., Wu, P., Lau, E. H. Y., and Cowling, B. J. (2022). Systematic review and meta-analyses of superspreading of SARS-CoV-2 infections. *Transboundary and Emerging Diseases*, 69(5).
5. **Liu, C.**<sup>†</sup>, Xu, L., Bai, Y., Xu, X., Lau, E. H. Y., Cowling, B. J., and Du, Z. (2022). Local surveillance of the COVID-19 outbreak. *Frontiers in Physics*, 10, 824369.

### Working Papers

1. **Liu, C.**, Murphy, C., Okoli, G. N., Sullivan, S. G., and Cowling, B. J. *Vaccine-induced severity attenuation and bias in test-negative estimates of vaccine effectiveness*. Manuscript in preparation.
2. **Liu, C.**, Cheung, J. K., Lin, Y., Wu, P., Wong, J. Y., and Cowling, B. J. *Hospital pressure, admission selection, and COVID-19 hospital fatality risk across epidemic periods in Hong Kong: a retrospective cohort study*. Manuscript in preparation.

## SELECTED PRESENTATIONS

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### EPIDEMICS<sup>10</sup>

San Diego, USA

Oral presentation

Dec 2025

*Impact of vaccine-induced severity attenuation on bias in vaccine-effectiveness estimates from test-negative designs.*

### EPIDEMICS<sup>9</sup>

Bologna, Italy

Poster presentation

Nov 2023

*Estimating the COVID-19 severity pyramid in Hong Kong, January 2020–January 2023.*

### Annual Scientific Meeting of the Hong Kong College of Community Medicine

Poster presentation

Hong Kong, China Sep 2022

*Surveillance for local and importation risk of COVID-19.*

## RESEARCH EXPERIENCE

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### Laboratory of Data Discovery for Health (D<sup>2</sup>4H)

Hong Kong SAR, China

Research Associate

Jul 2021–Jul 2022

- Supported data access, protocol development, and ethics approval for a Hospital Authority Data Collaboration Lab retrospective cohort study of oseltamivir effectiveness in hospitalized influenza patients using longitudinal clinical records.

- Co-developed an R package integrating EpiEstim and virosolver to simulate epidemic trajectories, estimate time-varying reproduction numbers, and support real-time monitoring of local COVID-19 transmission.
- Conducted systematic reviews and meta-analyses of SARS-CoV-2 transmissibility, synthesizing variant-specific reproduction numbers and superspreading parameters.

**Southern University of Science and Technology**  
*Research Assistant, Department of Statistics and Data Science*

Shenzhen, China  
Jul 2020–Jun 2021

- Conducted methodological literature reviews and developed estimation methods using the MM algorithm for bivariate binomial distributions and multivariate zero-inflated count data.

## TEACHING EXPERIENCE

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**The University of Hong Kong**  
CMED 6020 Advanced Statistical Methods I, *Tutor*  
CMED 6020 Advanced Statistical Methods I, *Teaching Assistant*

Spring 2025–Spring 2026  
Spring 2023–Spring 2024

**Southern University of Science and Technology**  
MA103A Linear Algebra A, *Teaching Assistant*  
MA102B Calculus A II, *Teaching Assistant*  
MA101B Calculus A I, *Teaching Assistant*

Fall 2019  
Spring 2019  
Fall 2018

## ADDITIONAL TRAINING

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**Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID)**  
*University of Washington, Department of Biostatistics*

July 2023

**Selected modules:** MCMC I and II for Infectious Diseases; Simulation-Based Inference for Epidemiological Dynamics; Statistics and Modeling with Novel Data Streams.

## TECHNICAL SKILLS

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**Programming:** R, Python; **Tools:** L<sup>A</sup>T<sub>E</sub>X, Git.

## LANGUAGES

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Mandarin (native); English (professional); Cantonese (conversational).